

Made Up of Cells Living things, or organisms, are made up of small, self-contained units called cells. Cells are the smallest units of an organism that can be considered alive. Cells can grow, respond to their surroundings, and reproduce. Despite their small size, cells are complex and highly organized.



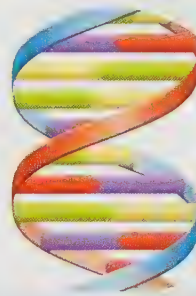
VIDEO

Learn about the characteristics of life and what it means to be alive.

Reproduction All organisms produce new organisms through a process called reproduction. There are two basic kinds of reproduction: sexual and asexual. The vast majority of multicellular organisms reproduce sexually. In **sexual reproduction**, cells from two different parents unite to produce the first cell of the new organism. In **asexual reproduction**, the new organism has a single parent.

Based on a Universal Genetic

Code Offspring usually resemble their parents. With asexual reproduction, offspring and their parents have the same traits. With sexual reproduction, offspring differ from their parents in some ways. Explaining how organisms inherit traits is one of the greatest achievements of modern biology. Biologists now know that the directions for inheritance are carried by a molecule called **DNA**. This genetic code, with a few minor variations, determines the inherited traits of every organism on Earth.



DNA helix

Growth and Development All living things grow during at least part of their lives. For some single-celled organisms, such as bacteria, growth is mostly a simple increase in size. Multicellular organisms, like this monarch butterfly, typically go through a process called development. During development, a single fertilized egg cell divides again and again to produce the many cells of mature organisms. As those cells divide, they change in shape and structure. This process is called differentiation, because it forms cells that look different from one another and perform different functions.

